Design and Evaluation of an Al-Based Behavioral and BNPL-Aware Credit Risk Prediction Model Using Transaction Data

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PROJECT ABSTRACT

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Abstract:

In this research, we create a credit scoring algorithm that is sensitive to BNPL activities utilizing a single set of data. This model uses data on spending behavior, credit use, BNPL activities, and liquidity to enhance credit risk assessment. For machine learning algorithms, Logistic Regression, Random Forest, XGBoost, and LightGBM are applied in predicting default rates. Finally, the results are converted to credit scores ranging from 300 to 900 through a process called FINSCORE.

Project Areas: Machine Learning and Credit Risk Management

Key Words: Credit Risk Prediction, Buy Now Pay Later (BNPL), Behavioral Credit Scoring, Transaction Data Analytics, Machine Learning, Default Prediction, Financial Behavior Modeling, Alternative Data, FinTech, LightGBM, XGBoost, Credit Score Modeling.

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Problem Statement

The conventional credit scoring models mostly focus on past credit repayment history and credit bureau-based information, which gives only a partial view of the consumer's financial behavior. With the emergence of digital financial services and Buy Now Pay Later (BNPL) products, there has been an unprecedented rise in the adoption of short-term credit transactions. Moreover, consumers are using services of different financial institutions, which makes it difficult for the lender to obtain a unified view of the consumer's financial behavior. The conventional credit evaluation models might not be considering other significant behavioral factors such as spending variability, BNPL dependency, liquidity issues, and cash flows. This project attempts to bridge this knowledge gap by conducting an empirical analysis of whether BNPL usage and unified multi-institution transaction data incorporation into behavioral credit scoring models can enhance credit risk classification relative to traditional credit scoring models.

Project Objectives

The major aim of this study is to empirically assess if incorporating Buy Now Pay Later behavioral measures and unified transaction data across multiple institutions can improve the effectiveness of credit risk assessment models.

The specific objectives of this study are as follows:

- To create a unified financial behavior data set by combining information relating to transaction-level spending patterns, BNPL, and account balance. This objective is concerned with creating a consolidated consumer financial profile that captures various financial behavior indicators, including volatility in spending, credit utilization, liquidity, and BNPL dependency.

- To create and implement credit scoring models that capture various consumer financial behaviors, including those utilizing machine learning techniques such as XGBoost, and LightGBM. These models will integrate traditional financial data and consumer financial behavior data to predict the probability of default in credit repayment.

- To empirically compare the effectiveness of credit scoring models that capture consumer financial behaviors with traditional credit risk models, utilizing various evaluation criteria including Accuracy, Precision, Recall, F1 Score, and ROC-AUC. This objective aims to establish whether the addition of BNPL and consumer financial behavior data improves the accuracy of credit risk classification models.

- To create a BNPL-aware credit scoring model (FINSCORE), which illustrates the ability to create a dynamic credit score that captures consumer financial health and repayment risk, utilizing a unified financial behavior data set.

Project Scope

The objective of this project is to empirically investigate the significance of using Buy Now Pay Later (BNPL) behavioral indicators and unified multi-institution transactional data for enhancing consumer credit risk assessment. The specific focus of the study is to develop and evaluate a behavioral credit scoring system that includes alternative financial information, such as spending patterns, BNPL, liquidity, and credit utilization.

In the context of this project, the following components will be included:

- Analysis of existing credit scoring methods and the identification of shortcomings in the absence of behavioral transaction data and BNPL-related financial activities.

- Gathering and processing financial transaction data sets created using simulated banking data to reflect consumer spending behavior, BNPL activity, and account balance movements.

- Building a comprehensive financial profile for individual consumers by consolidating transaction data from multiple simulated financial accounts.

- Extraction and development of behavioral financial features such as spending variability, BNPL dependency, credit utilization, liquidity stress, and cash flow stability.

- Building behavioral credit scoring models using machine learning algorithms such as Logistic Regression, Random Forest, XGBoost, and LightGBM.

- Empirical analysis of model performance by comparing the behavioral credit scoring models to baseline models using performance metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.

- Presentation of a BNPL-aware behavioral credit scoring system (FINSCORE) to demonstrate how combined transaction-level financial information can enhance credit risk categorization.

The scope of the project is constrained by the following considerations:

- The empirical study will be performed using simulated or publicly available data and not using any banking or consumer financial information that is proprietary in nature.

- The project is limited to the credit risk assessment phase and does not involve the construction of a full-fledged digital lending platform.

- Compliance with regulations, loan management, collection management, pricing algorithms, and risk management of portfolios are out of the scope of the current study.

- The study will not involve any integration with live banking systems, credit bureaus, or open banking infrastructure such as the Account Aggregator system.

The scope of the project is designed to ensure that the study is practical and can be completed within the project period and allows for an effective empirical study of the impact of behavioral financial information and BNPL in contemporary credit risk assessment.

Methodology

The current research utilizes a quantitative empirical research methodology to assess whether behavioral financial indicators based on transaction-level data and Buy Now Pay Later (BNPL) transactions improve credit risk assessment relative to traditional credit score-based financial indicators.

Research Design

The research follows a data-driven model development and evaluation research design. In this research, a unified consumer financial data set is built using simulated transaction-level banking data, and behavioral credit score models are developed and evaluated.

Research Type

The research type used in this research is applied empirical research, which deals with model development and evaluation for predictive models used for credit risk classification based on behavioral financial indicators.

Population and Sample Framework

The population considered in this research deals with retail bank customers who participate in digital financial transactions such as debit card transactions, credit card transactions, and Buy Now Pay Later transactions.

Due to the lack of access to publicly available unified transaction-level data from multiple financial institutions, this research utilizes synthetic financial transaction data generated to mimic realistic financial transaction data for consumers. This data set comprises approximately 12,000 simulated consumers with transaction history on various financial activities such as spending, Buy Now Pay Later transactions, credit card transactions, and balance changes.

Each consumer profile includes variables representing:

- Demographic attributes (age, income)
- Credit characteristics (credit cards, credit limits)
- Spending patterns
- BNPL exposure
- Cash-flow indicators
- Liquidity conditions

Data Generation and Preparation

A synthetic financial data set is created to mimic real-world financial activities in various financial accounts for individuals. Transaction-level data include consumer spending on various categories such as groceries, utility bills, travel expenses, and retail purchases, and BNPL installment payments and credit card transactions.

The transaction-level data is aggregated to obtain a unified financial profile for each individual consumer. Derived behavioral features include metrics such as spending variability, BNPL adoption rate, credit utilization rate, liquidity stress, and minimum account balance.

Model Development

The paper proposes developing behavioral credit scoring models based on supervised machine learning algorithms. The models used in this paper are:

- XGBoost
- LightGBM

These models are used to predict the probability of default based on both traditional financial variables and newly proposed behavioral variables.

Model Evaluation

The performance of the models is evaluated based on various metrics used to evaluate the performance of classification models, which are:

- Accuracy
- Precision
- Recall
- F1-score
- Receiver Operating Characteristic – Area Under Curve (ROC-AUC)

The models are also used to perform comparative analysis to see whether there is an improvement in model performance by including variables related to BNPL and transaction-level financial activities.

Framework Demonstration

Based on the empirical results, the study demonstrates a BNPL-aware behavioral credit scoring framework referred to as **FINSCORE**, which illustrates how unified transaction data can be incorporated into modern credit evaluation systems.

Plan vs Progress

Milestone	Timeline	Deliverable	Status	Completion
Topic finalization & proposal submission	Jan 2026	Project outline	Completed	100%
Literature review on BNPL and behavioral credit scoring	Jan – Feb 2026	Literature summary	Completed	100%
Synthetic transaction dataset generation	Feb 2026	Customers, transactions, BNPL dataset	Completed	100%
Feature engineering & behavioral variable creation	Feb – Mar 2026	Unified dataset	Completed	100%
Model development (Logistic, RF, XGBoost, LightGBM)	Mar 2026	Credit risk prediction models	Completed	100%
Model evaluation & benchmarking	Apr 2026	Performance comparison	Completed	100%
Development of FINSCORE framework	Apr 2026	Behavioral scoring model	Completed	100%
Final report preparation	Apr – May 2026	Dissertation submission	Completed	100%

Table 1: Plan vs Progress

Detailed description of the project

1. Literature Review

- **Introduction**

The last decade has seen tremendous growth in the financial technology industry, which has revolutionized the way consumers access various financial services. One of the most prominent innovations to have come out of this revolution is the concept of Buy Now Pay Later (BNPL) services. BNPL services offer consumers the opportunity to purchase products or services immediately while paying in installments in the short term, usually without any interest or minimal interest charged. Unlike other credit facilities, BNPL services offer consumers credit facilities with minimal credit checks, which makes these services extremely popular among young consumers and individuals with poor credit history.

The popularity of BNPL services has increased manifold in recent times across the globe, especially in the context of online shopping platforms. Companies like Klarna, Afterpay, Affirm, and Paytm Postpaid are prominent BNPL companies that have seamlessly integrated into the online shopping ecosystem, offering consumers instant credit facilities while shopping online. These services have become extremely popular, with millions of consumers who were earlier dependent on credit cards or personal loans to opt for BNPL services while shopping online.

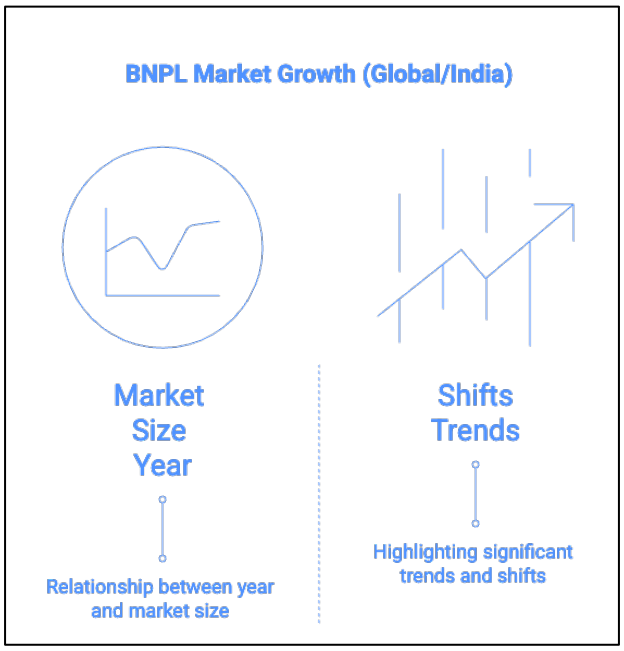


Fig 1: BNPL Market Growth

Although BNPL services provide financial accessibility and convenience for consumers, it has also created new patterns of financial behavior that are not included in traditional credit scoring. Consumers may have multiple BNPL obligations at a given time using various platforms. This may increase the risk of financial behavior. BNPL transactions occur outside traditional financial institutions. These obligations may not be included in traditional credit scoring. This may not provide a complete picture of credit risk assessment.

Traditionally, credit scoring has been designed to assess long-term credit behavior, including mortgage loans, credit cards, and personal loans. However, with the evolution of digital payment systems and transaction-level financial behavior, it has created new possibilities for analyzing financial behavior. This research aims to understand the role of BNPL-related behavioral data in credit risk prediction. Machine learning techniques may provide a more accurate credit risk prediction compared to traditional credit scoring.

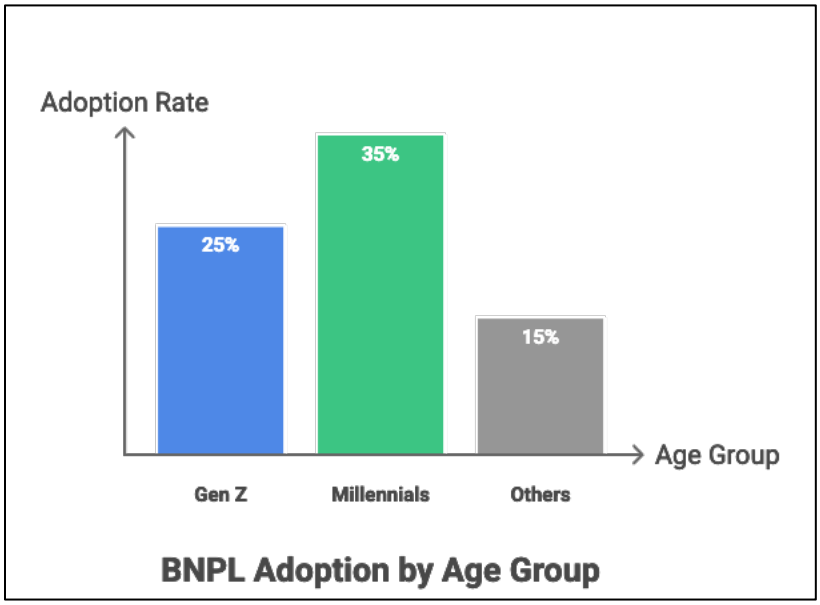


Fig 2: BNPL Adoption by Age Group

- **Background and Motivation**

The role of credit scoring in the financial system

The credit scoring technique plays an important role in the financial system because it helps lenders to determine the probability of getting loans back. It helps to predict credit risk more accurately so that default risk can be reduced.

Traditional credit scoring techniques

Traditional credit scoring techniques like FICO scores and credit bureau scores have been used by banks and other financial institutions for several decades. These techniques consider several important factors while determining credit scores, including payment history, credit utilization ratio, credit history, credit type, and credit inquiries in recent times. These factors are important in understanding consumer behavior in the long run, but these factors cannot be used to understand recent trends in consumer behavior.

The digitalization of the financial industry

The digitalization of the financial industry has led to the generation of behavioral finance data in recent times. Consumers use digital platforms like mobile banking, payment systems, and digital wallets to perform most of their financial transactions these days. It is now possible to analyze consumer spending behavior in much more detail.

The BNPL services form a distinct segment in digital financial services. In contrast to other credit instruments, BNPL transactions are frequently integrated directly into retail platforms and involve multiple installment payments within a short period. Consumers frequently use BNPL services to make purchases, thereby creating a pattern of recurring micro-credit debt.

From a risk management viewpoint, BNPL services generate novel credit risk signals. For instance, consumers frequently using BNPL services to make discretionary purchases may be under significant financial stress or exhibit impulsive spending behavior. In addition, consumers who accumulate multiple BNPL installment plans on different platforms may be more likely to default on repayments.

However, credit scoring models generally do not have visibility into these behavioral signals. Consequently, credit institutions using traditional credit scoring models may not be able to assess the associated financial risk posed by BNPL consumers.

The rationale for this study is to examine the possibility of incorporating BNPL-related behavioral factors into credit risk prediction models to enhance the predictability of such models. With the aid of transactional data and ML methods, it may be possible to create more holistic credit scoring models that reflect contemporary financial behaviors exhibited by consumers.

Stakeholder Perspective

- Banks → risk reduction
- Regulators → transparency, compliance
- Consumers → fair scoring
- BNPL providers → data sharing concerns

- **Limitations of Traditional Credit Scoring Models**

Conventional credit scoring models have been effective creditworthiness evaluation tools in conventional credit environments. However, conventional credit scoring models were developed in a financial environment where transactions were conducted primarily through bank-based credit institutions. With the evolution of the financial environment, the following limitations of conventional credit scoring models have become more apparent.

3.1 Dependence on Historical Credit Information

Conventional scoring models heavily depend on past credit information, such as loan payment history and credit card behavior. Although this information is valuable, it does not reflect recent changes in the financial status of a consumer.

3.2 Inadequate Coverage of Thin-File Consumers

Some consumers, especially young adults or those living in developing markets, lack a credit history. These consumers are termed thin-file consumers. Conventional credit scoring models will not be able to evaluate the creditworthiness of such consumers due to their reliance on a credit history.

3.3 Inadequate Coverage of Behavioral Financial Information

Conventional credit scoring models use aggregate financial information rather than behavioral information. For example, a credit scoring model will use total debt but will not account for transaction frequency. This inadequate information will not enable the model to identify new financial risk patterns.

3.4 Lack of BNPL Data Integration

One issue associated with BNPL services is that they are not part of the traditional banking structure, and in most cases, BNPL companies do not report consumer transactions to credit reporting agencies. This means that when a consumer applies for credit, the lender will not be able to see their BNPL obligations.

The above challenges emphasize the importance of developing other means of credit risk assessment, which involve behavioral finance and digital transactions.

- **Alternative Data in Credit Risk Modeling**

In order to overcome the shortcomings associated with the traditional credit score prediction models, researchers and financial institutions have been actively looking to incorporate alternative data in credit risk prediction models. Alternative data refers to any non-traditional source of information that can be used to obtain knowledge about consumer financial behavior.

Examples of alternative data include:

- Transaction-level banking data
- Mobile payment activity
- E-commerce purchase records
- Utility and telecom payment history
- Digital wallet usage
- Social and behavioral data

Transaction-level financial information is especially valuable because it contains detailed information regarding consumer spending habits. For example, transaction-level information can be used to determine spending frequency, volatility, merchant type, and seasonal spending patterns.

Analyzing this type of information can be used to identify behavioral patterns that can be indicative of potential financial stress or instability. For example, increased spending levels, overdrafts, or income volatility can be indicative of higher credit risk.

The use of BNPL transaction information is a unique form of alternative financial information because BNPL transactions are generally frequently occurring and involve low dollar amounts. However, these obligations can become significant levels of debt when aggregated across multiple transactions.

The use of BNPL transaction information in credit scoring models can be used to identify additional signals not available in traditional credit scoring models.

- **Machine Learning in Credit Risk Prediction**

In recent years, machine learning approaches have received considerable interest in the field of credit risk modeling. This is because machine learning models have the ability to discover complex relationships in large datasets, which are not linear in nature.

Several machine learning algorithms have demonstrated strong performance in credit risk prediction tasks, including:

- Random Forest
- Gradient Boosting Machines (GBM)
- XGBoost
- Support Vector Machines (SVM)
- Neural Networks

5.1 Random Forest

Random Forest is a type of ensemble learning method that uses multiple decision trees to generate predictions. This method is helpful in overcoming issues such as overfitting, which can result in improved predictions.

5.2 Gradient Boosting

Gradient Boosting is a type of machine learning method that uses ensemble learning to develop multiple models, where each model attempts to rectify the predictions of all other models. This type of learning is helpful in addressing complex issues within data.

5.3 XGBoost

XGBoost is a type of machine learning method that is derived from extreme gradient boosting, which is helpful in solving predictive modeling challenges, especially in finance, due to its high performance.

Machine learning models have shown promise in analyzing financial data, especially those related to transactions, due to their ability to process multiple features, where even minor changes in data can help predict patterns that would not be visible through traditional statistical methods.

Ethical Considerations in ML Credit Scoring

Include:

- Algorithmic bias (e.g., income proxies)
- Transparency (XGBoost = black box)
- Data privacy concerns
- Risk of financial exclusion

- **Research Gap**

Despite the significant progress made in the application of machine learning for credit risk prediction, there are still some gaps in the existing literature that this study aims to address.

The existing literature has concentrated mostly on data sets that are collected from individual financial institutions. For instance, the data sets could have been collected from individual banks or credit card companies.

However, the current financial behavior of consumers involves interacting with more than one financial institution. For instance, the consumers may use more than one payment service, BNPL services, and other digital wallets.

Additionally, the existing literature has not extensively addressed the impact of BNPL behavioral data on credit risk prediction. Given the current growth rate of BNPL services, the impact of BNPL transactions on credit risk prediction is an important area of research.

The current study aims to bridge the existing gaps by examining the impact of BNPL behavioral data on credit scoring.

- **Behavioral and Interdisciplinary Perspective**

  - **Psychology:** impulsive spending, instant gratification (BNPL → buy-now mindset)
  - **Sociology:** peer influence, lifestyle consumption, social pressure
  - **Economics:** consumption smoothing, liquidity constraints

- **Growth of BNPL Industry and Market Significance**

The international BNPL industry has experienced exponential growth in the last few years. The size of the BNPL market was above USD 120 billion in 2023, and this market is estimated to increase by more than 20% CAGR in upcoming years. The usage of BNPL services has been growing rapidly in India due to high digital transactions, and players like Paytm Postpaid and LazyPay have played an important role in shaping the sector.

The majority of BNPL users comprise young people in the age bracket of 18 to 35 years, most of whom do not have any experience or records of using any credit products before. Reports claim that around 40% of BNPL users do not even own a single credit card, which indicates the need for alternative methods to score customers' creditworthiness.

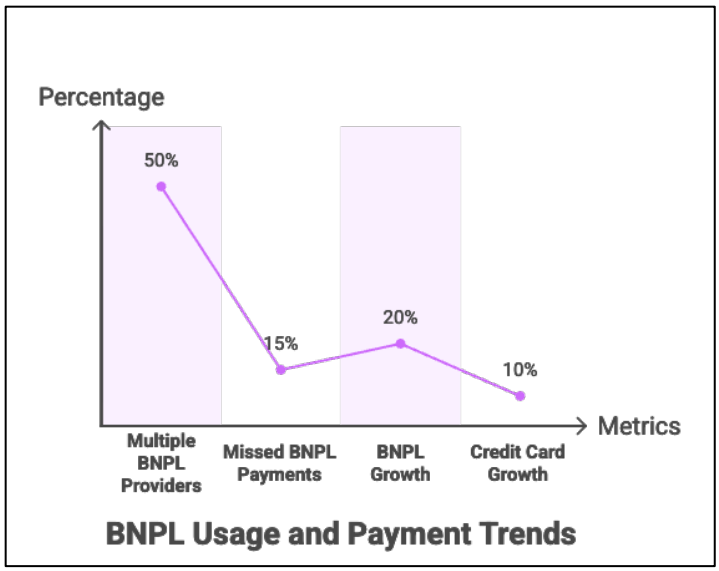


Fig 3: BNPL Usage and Payment Trends

Nevertheless, an increasing number of BNPL loans have resulted in overleverage and hidden debts among customers as multiple BNPL loans go unnoticed by traditional credit bureaus.

- **Case Study: BNPL Risk and Credit Blind Spots**

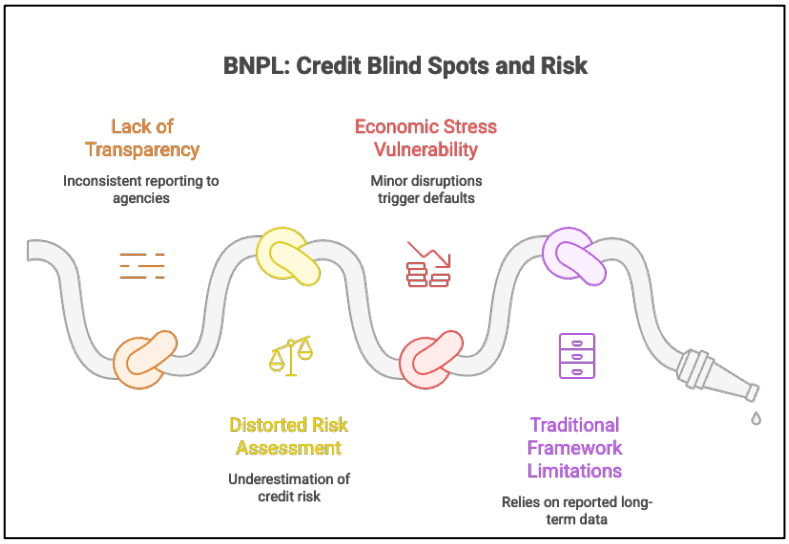


Fig 4: BNPL Blind Spot Case Study

- One such case study can be observed in the United States and Australian markets, where BNPL services have experienced rapid adoption through major providers such as Klarna and Afterpay. These platforms have seamlessly integrated with e-commerce ecosystems, allowing consumers to split payments into multiple short-term installments with minimal friction and limited credit checks.
- Recent studies and industry reports indicate that a significant proportion of BNPL users maintain multiple concurrent installments plans across different providers. This fragmented borrowing behavior creates a scenario where individual financial obligations are distributed across platforms, making it difficult for any single lender or credit bureau to obtain a complete view of a borrower's liabilities. Since BNPL transactions are often not consistently reported to traditional credit reporting agencies, this leads to a lack of transparency in the borrower's overall credit exposure.
- For instance, a consumer may appear financially stable when evaluated using conventional credit metrics such as repayment history, credit utilization, and outstanding loans. However, the same individual may simultaneously be servicing several BNPL installments across different platforms. This hidden layer of short-term debt can significantly strain liquidity, especially when combined with regular expenses, increasing the likelihood of missed payments or default.
- Empirical observations from these markets suggest that such "invisible debt accumulation" can distort risk assessment models, leading to an underestimation of credit risk. This issue becomes particularly critical during periods of economic stress, where even minor disruptions in income can trigger cascading defaults across multiple BNPL obligations.

- This case highlights a fundamental limitation of traditional credit scoring frameworks, which rely heavily on reported and long-term credit data. It underscores the necessity of integrating BNPL-related behavioral indicators and transaction-level insights into modern credit risk assessment models to ensure a more comprehensive and accurate evaluation of consumer financial health.
- India → under-regulated, high growth

  US → emerging regulation

  Australia → stricter BNPL oversight

- **Research Objectives**

The primary objective of this research is to evaluate the impact of BNPL behavioral data on credit risk prediction models.

The specific objectives include:

1. To analyze consumer financial behavior using transaction-level data that includes BNPL activity.
2. To identify key behavioral factors associated with BNPL usage.
3. To develop credit risk prediction models using machine learning algorithms.
4. To compare the performance of models with and without BNPL-related features.
5. To determine whether BNPL behavioral data improves prediction accuracy.

- **Stages**

The research methodology involves several stages.

8.1 Data Collection

The dataset used in this study includes transaction-level financial records from multiple institutions. The dataset contains information about consumer purchases, BNPL transactions, repayment activity, and other financial behaviors.

8.2 Feature Engineering

Relevant behavioral features are extracted from the raw transaction data, such as:

- BNPL usage frequency
- Average installment value
- Number of active BNPL plans
- Spending volatility
- Merchant category distribution
- Transaction frequency

8.3 Model Development

Several machine learning algorithms will be implemented to predict credit risk, including Random Forest, Gradient Boosting, and XGBoost.

8.4 Model Evaluation

The models will be evaluated using standard classification metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Area Under the ROC Curve (AUC)

Comparisons will be made between models trained with traditional credit features and models that incorporate BNPL behavioral data.

- **Expected Contributions**

This research aims to contribute to the field of financial analytics in several ways.

First, it provides insights into how BNPL usage patterns influence consumer credit risk. Second, it demonstrates the value of transaction-level behavioral data in credit scoring models. Third, it evaluates the effectiveness of machine learning algorithms in integrating alternative financial data sources.

The findings of this research may be useful for financial institutions, fintech companies, and policymakers seeking to develop more accurate and inclusive credit assessment frameworks.

- **Summary**

The growth of BNPL has resulted in new forms of financial behavior that are difficult to predict through traditional credit risk assessment models. BNPL has provided greater financial accessibility and convenience; however, it has also led to new forms of credit exposure that may not be addressed through traditional credit scoring models.

Conventional credit scoring models rely on credit history to make predictions; hence, it is difficult to predict behavioral financial patterns through such models.

Recent advancements in machine learning and alternative data analytics provide an opportunity to improve credit risk prediction through detailed financial behavior data. BNPL financial behavior can help predict credit risks more effectively through financial data analytics.

This study aims to examine the potential of BNPL financial behavior in credit scoring models and determine whether BNPL can help improve credit scoring models. As financial services become more digital, it is important to develop more comprehensive credit risk models to predict financial risks effectively.

2. Dataset Generation

The dataset contains approximately **12,000 consumers**, each with transaction-level financial records spanning three years of financial activity.

The generated dataset includes three primary tables:

Dataset	Description
Customers	Demographic and credit profile
Transactions	Transaction-level financial activity
BNPL Installments	BNPL purchase repayment records

Table 2: Dataset Description

The transaction dataset includes variables such as:

- transaction date
- transaction amount
- payment channel (debit card, credit card, BNPL)
- spending category
- account balance after transaction

This structure allows the project to simulate realistic financial behavior including:

- spending variability
- credit card usage
- BNPL purchases
- liquidity stress

3. Feature Engineering

3.1 Introduction

Feature engineering is an important aspect of the development of machine learning models. In the context of credit risk prediction, feature engineering is vital. The success of the prediction model depends on the quality and relevance of the feature sets. In this study, feature engineering is the focal point. Feature engineering transforms the raw financial data obtained at the transactional level into relevant feature sets that are representative of the behavior and financial risks of the consumers.

With the advent of modern financial ecosystems and the inclusion of the Buy Now Pay Later (BNPL) option in the financial system, the traditional financial feature sets are no longer representative of the creditworthiness of the consumers. This study highlights the importance of developing a unified financial profile for the consumers by incorporating the transactional data and the traditional feature sets. The feature sets are engineered in such a way that they are representative of the behavioral patterns of the consumers. The feature sets are representative of the static financial characteristics as well as the dynamic behavioral characteristics.

3.2 Data Aggregation and Unified Financial Profile

The data used in this study is in a raw form and comprises transaction-level data from various financial platforms such as banking and BNPL transactions. Each transaction contains various attributes such as transaction amount, date and time, merchant category, and type of transaction.

However, in order to use this data for predictive modeling techniques, it is aggregated based on individual consumers. Aggregation essentially means summarizing various types of information into a set of meaningful indicators that represent financial behavior.

The process of aggregating this data involves:

- Grouping various types of transaction data based on individual consumers
- Calculating aggregate values such as total, average, and variance values
- Identifying patterns in data based on a specific period of time, such as monthly or yearly trends
- Combining BNPL-related data with financial transaction data

This gives rise to a unified financial profile in which each row represents a consumer and each column represents a feature.

3.3 Key Financial Indicators

The following key variables were derived from the aggregated dataset to capture essential aspects of consumer financial behavior:

Variable	Description
Total Spend	Annual spending volume
Average Spend	Mean transaction size
Spend Volatility	Variability in spending behavior
BNPL Total	Total BNPL purchase value
BNPL Transactions	Number of BNPL purchases
Credit Utilization	Ratio of spending to credit limit
Minimum Balance	Lowest account balance observed

Table 3: Key Financial Indicators

Each of these variables provides unique insights into different dimensions of financial behavior.

3.4 Explanation of Core Features

3.4.1 Total Spend

Total Spend represents the aggregate amount of money spent by a consumer over a specified period, typically one year. This variable reflects the overall level of financial activity and consumption.

Higher spending levels may indicate greater purchasing power; however, excessive spending relative to income or credit limits may signal financial stress. Therefore, Total Spend must be interpreted in conjunction with other variables such as credit utilization and income proxies.

3.4.2 Average Spend

Average Spend is calculated as the mean transaction amount. It provides insights into the typical size of purchases made by the consumer.

A higher average transaction value may indicate fewer but larger purchases, while a lower average may reflect frequent small transactions. When combined with transaction frequency, this variable helps distinguish between different spending patterns.

3.4.3 Spend Volatility

Spend Volatility measures the variability in transaction amounts over time, typically calculated using statistical measures such as standard deviation or variance.

High volatility may indicate inconsistent financial behavior, which could be associated with income instability or impulsive spending. Such patterns are often linked to higher credit risk.

Conversely, low volatility suggests stable and predictable spending behavior, which is generally associated with lower financial risk.

3.4.4 BNPL Total

BNPL Total represents the cumulative value of purchases made using BNPL services.

This variable captures the extent to which a consumer relies on short-term financing for purchases. A high BNPL Total may indicate increased dependency on deferred payments, which could be a sign of financial strain.

3.4.5 BNPL Transactions

BNPL Transactions refer to the number of purchases made using BNPL services.

This variable reflects the frequency of BNPL usage. Frequent use of BNPL for everyday expenses may indicate reliance on credit for routine consumption, which can be a potential risk factor.

3.4.6 Credit Utilization

Credit Utilization is defined as the ratio of total spending to the available credit limit.

This is a widely used metric in traditional credit scoring models and serves as an important indicator of financial health. High credit utilization suggests that a consumer is using a large portion of their available credit, which may increase the likelihood of default.

3.4.7 Minimum Balance

Minimum Balance represents the lowest account balance observed over a given period.

This variable provides insights into liquidity and financial resilience. A consistently low or negative balance may indicate financial vulnerability and a higher risk of default.

4. Behavioral Variables

Behavioral indicators were derived to capture patterns associated with financial instability:

Examples include:

- BNPL Income Ratio
- Credit Utilization Ratio
- Spend-to-Income Ratio
- Cash Advance Ratio
- External Financial Support Ratio

These indicators help detect patterns such as:

- BNPL dependency
- liquidity stress
- high credit utilization
- unstable spending behavior

5. Preliminary Model Development

5.1 Introduction

The development of predictive models is an important part of this research study since it facilitates the quantitative measurement of the credit risk faced by consumers. In the context of the changing financial environment, especially with the advent of BNPL services, it becomes important to examine the potential effectiveness of advanced machine learning approaches in identifying the changing patterns of credit behavior.

The current section highlights the preliminary development of predictive models with the objective of measuring the likelihood of default in the context of consumer credit. The objective of developing such predictive models is to examine the effectiveness of the incorporation of behavioral factors in the context of transactional data and BNPL services.

To accomplish this objective, two state-of-the-art machine learning algorithms have been implemented: Extreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM). These algorithms have been chosen for their high performance in classification problems for structured data and for their ability to effectively deal with complex relationships between variables.

5.2 Model Selection Rationale

The choice of machine learning models is an important factor that influences the success of the credit risk prediction. For example, statistical models, e.g., logistic regression, are popular for use in credit scoring. This is because of their simplicity and interpretability. However, statistical models are weak in handling non-linear relationships, especially when dealing with high-dimensional behavioral data.

On the other hand, the use of ensemble learning models, e.g., XGBoost and LightGBM, is suitable for this study for the following reasons:

5.2.1 Ability to Capture Non-linear Relationships

Consumer financial behavior is complex and has non-linear characteristics. For example, an increase in spending may not necessarily imply a risky behavior. On the other hand, an abnormal increase in the use of BNPL services and irregular repayment may imply financial difficulties. Gradient boosting has the capability of capturing non-linear relationships.

5.2.2 Robustness to Feature Interactions

The behavioral characteristics obtained from transactional data may also have feature interactions. For example, spending volatility and transaction frequency may have an impact on the use of BNPL services. Ensemble learning has the capability of capturing feature interactions and using this information for prediction.

5.2.3 Handling of Large and High-Dimensional Data

The transactional data has a high number of features. Both XGBoost and LightGBM are efficient in handling high-dimensional data. Therefore, they are appropriate for this problem domain.

5.2.4 Proven Performance in Credit Risk Modeling

Several studies have shown the superiority of gradient boosting algorithms compared to other algorithms for credit risk prediction. The capability of the gradient boosting algorithm to reduce the prediction error by learning from the data makes it appropriate for classification problems.

5.3 Overview of Implemented Models

5.3.1 XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced implementation of gradient boosting that uses decision trees as base learners. It builds models sequentially, where each new tree attempts to correct the errors made by the previous trees.

Key characteristics of XGBoost include:

- Regularization to prevent overfitting
- Efficient handling of missing values
- Parallel processing capabilities
- High predictive accuracy

The model minimizes a loss function using gradient descent, while also incorporating regularization terms to control model complexity.

5.3.2 LightGBM

LightGBM (Light Gradient Boosting Machine) is another gradient boosting framework designed for high efficiency and scalability. It differs from traditional boosting methods in its tree-building approach.

Key features of LightGBM include:

- Leaf-wise tree growth strategy
- Faster training speed compared to XGBoost
- Lower memory consumption
- Ability to handle large datasets

LightGBM is particularly effective when dealing with large-scale data and can achieve comparable or better performance than XGBoost in certain scenarios.

5.4 Data Preparation and Feature Integration

The predictive models are trained using a combination of traditional financial variables and newly engineered behavioral indicators. Proper data preparation is essential to ensure that the models can effectively learn from the data.

5.4.1 Traditional Financial Variables

These variables represent conventional credit-related information typically used in credit scoring models, including:

- Credit utilization ratio
- Outstanding debt
- Payment history
- Number of credit accounts
- Length of credit history

These features serve as the baseline inputs for the models.

5.4.2 Behavioral Financial Indicators

Behavioral indicators are derived from transaction-level data and BNPL usage patterns. These features capture dynamic aspects of consumer financial behavior, such as:

- Frequency of BNPL transactions
- Average installment size
- Number of active BNPL plans
- Spending volatility (variance in transaction amounts)
- Transaction frequency over time
- Merchant category diversity

These variables provide insights into consumer spending habits, financial discipline, and potential risk exposure.

5.4.3 Data Preprocessing

Before model training, the dataset undergoes several preprocessing steps:

- **Handling Missing Values:** Missing data is imputed using appropriate techniques such as mean imputation or model-based imputation.
- **Feature Scaling:** Although tree-based models are less sensitive to scaling, normalization may still be applied for consistency.
- **Encoding Categorical Variables:** Categorical features are converted into numerical representations using techniques such as one-hot encoding or label encoding.
- **Train-Test Split:** The dataset is divided into training and testing subsets to evaluate model performance on unseen data.

5.5 Model Training Process

The process of training the machine learning models involves feeding the data into the machine learning algorithms and optimizing the parameters of the models for the best performance.

5.5.1 Training Procedure

- Training data is used in the process.
- Hyperparameters are tuned using grid search, random search, etc.
- Cross-validation is used for the robustness of the models, preventing overfitting, etc.

5.5.2 Hyperparameter Tuning

Some important hyperparameters are tuned for the XGBoost algorithm, as well as the LightGBM algorithm, which are as follows:

- Learning rate
- Maximum depth
- Number of estimators
- Subsampling ratio
- Regularization
- It is important that the hyperparameters are tuned for the best performance of the models.

5.5.3 Handling Class Imbalance

In credit risk prediction, the number of default cases is generally less compared to the number of non-default cases, so class imbalance occurs in the data, which needs to be handled as follows:

- Resampling
- Class weights
- Use of evaluation metrics

5.6 Model Evaluation Metrics

To assess the performance of the predictive models, several classification metrics are used. Each metric provides a different perspective on model effectiveness.

5.6.1 Accuracy

Accuracy measures the proportion of correctly classified instances out of the total number of observations.
While accuracy is easy to interpret, it may not be sufficient in imbalanced datasets, as a model can achieve high accuracy by simply predicting the majority class.

5.6.2 Precision

Precision measures the proportion of true positive predictions among all predicted positives.
In the context of credit risk, precision indicates how many predicted defaulters actually defaulted. High precision is important to avoid falsely labeling low-risk customers as high-risk.

5.6.3 Recall

Recall, also known as sensitivity, measures the proportion of actual positive cases that are correctly identified by the model.
In credit risk prediction, recall is crucial because failing to identify a defaulter can result in financial losses for lenders.

5.6.4 F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of model performance, especially in cases of class imbalance.

5.6.5 ROC-AUC

The Receiver Operating Characteristic (ROC) curve plots the true positive rate against the false positive rate at different threshold levels. The Area Under the Curve (AUC) provides a summary measure of the model's ability to distinguish between classes.
A higher ROC-AUC value indicates better model performance in distinguishing between defaulters and non-defaulters.

5.7 Comparative Evaluation Framework

To determine the effectiveness of behavioral financial indicators, two sets of models are developed:

1. **Baseline Models:**
   Trained using only traditional financial variables.
2. **Enhanced Models:**
   Trained using both traditional variables and behavioral indicators, including BNPL-related features.

The performance of these models is compared using the evaluation metrics described above.

5.8 Expected Outcomes

It is hypothesized that models incorporating behavioral financial indicators will outperform baseline models in terms of predictive accuracy and overall performance.

Specifically, the inclusion of BNPL-related features is expected to:

- Improve detection of high-risk consumers
- Capture emerging financial stress patterns
- Enhance model sensitivity (recall)
- Provide a more comprehensive understanding of credit behavior

5.9 Implications for Credit Risk Assessment

If the results confirm that behavioral indicators improve model performance, this would have significant implications for credit risk assessment practices.

Financial institutions may benefit from:

- More accurate credit scoring models
- Improved risk management strategies
- Enhanced ability to serve underbanked populations
- Better integration of fintech data sources

Furthermore, incorporating BNPL data into credit scoring frameworks could help address gaps in traditional credit systems and provide a more holistic view of consumer financial health.

5.10 Summary

The preliminary model development phase creates a platform to assess the impact of behavioral financial data on credit risk predictions. Leveraging sophisticated machine learning techniques such as XGBoost and LightGBM, this study attempts to identify intricate patterns in consumer financial behavior, which conventional credit scoring models do not account for.

Applying multiple evaluation metrics ensures a wide scope of assessing the performance of the models, especially considering class imbalance issues. Comparing the performance of baseline models with improved versions is expected to shed light on the benefits of incorporating BNPL-based behavioral factors.

This study is a contribution to the increasing body of literature on alternative data-driven credit risk modeling, emphasizing the importance of machine learning techniques in enhancing financial decision-making processes in the age of technological advancements.

6. Cost-Benefit Analysis of BNPL-Aware Credit Scoring Model

Component	Estimated Cost
Data Infrastructure	₹5–10 Lakhs
Model Development	₹3–5 Lakhs
Integration/API Systems	₹5–8 Lakhs
Compliance & Security	₹2–4 Lakhs
Total Estimated Cost	**₹15–25 Lakhs**

Table 4: Development Cost Estimate

Benefit	Impact
Reduction in Default Rate	10–20%
Improved Risk Detection	High-risk users identified earlier
Better Lending Decisions	Increased approval accuracy
Inclusion of Thin-file Users	+15–25% new customers

Table 5: Benefits Quantified

6.1 Financial Impact Example

If a bank has:

- ₹100 Cr loan portfolio
- 5% default rate

Default Loss = ₹5 Cr

If model reduces default by **15%**:

Savings = ₹75 Lakhs annually

Model pays for itself in < 1 year

6.2 Regulatory Considerations

Implementation of such a model must comply with:

- RBI Digital Lending Guidelines
- Data Privacy Laws (DPDP Act, GDPR equivalent)
- Consent-based data sharing (Account Aggregator framework)
- Fair lending & explainability requirements

Non-compliance risks:

- Regulatory penalties
- Model rejection by banks
- Legal exposure

6.3 Long-Term Economic Viability

The long-term sustainability of the proposed model depends on:

- Increasing digital transaction penetration
- Growth of BNPL ecosystem
- Integration with open banking frameworks
- Declining marginal cost of data processing

Over time, the model becomes:

- More accurate (more data)
- Cheaper to run
- More scalable

6.4 Implementation Challenges

Operational Challenges:

- Data integration across institutions
- Data standardization issues
- Real-time processing complexity
- Model deployment latency

Regulatory Challenges:

- Data sharing restrictions
- Consent management
- Explainability requirements (black-box models issue)
- Bias & fairness concerns

Resource Management

Resource	Requirement	Availability
Hardware	Laptop with Python environment	Available
Software	Python, Jupyter Notebook, VS Code	Available
Data	Banking transaction dataset	Generated
Libraries	Scikit-learn, XGBoost, LightGBM, Streamlit	Available

Table 6: Resource Requirements

The project extensively utilizes Python libraries and frameworks for data processing, modeling, and visualization. Each library is intended for a specific objective in the entire workflow:

Scikit-learn:

It is primarily utilized for the implementation of basic machine learning algorithms, data preprocessing, and performance measurement criteria such as accuracy, precision, recall, and F1-score. It also includes other utility functions, such as splitting the dataset into a training set and a test set, as well as cross-validation.

XGBoost:

It is a high-performance gradient boosting library that is utilized for building predictive models. This library is particularly useful for dealing with structured financial data and modeling non-linear relationships in credit scores.

LightGBM:

It is a gradient boosting library that is highly efficient, particularly for large datasets. This library is useful for building models, as it is faster than traditional boosting algorithms.

Pandas and NumPy:

These libraries are crucial for data processing and mathematical computations. Pandas is utilized for managing transaction data, while NumPy is useful for mathematical computations.

Matplotlib and Seaborn:

These libraries are utilized for data visualization. These libraries are particularly useful for understanding patterns, trends, and relationships in financial behavior.

Streamlit:

Streamlit is used to create an interactive web-based dashboard for the project. It allows visualization of model predictions, user inputs, and model performance metrics in a user-friendly way. This can also be helpful in showing how the credit risk model can be applied in real-world scenarios, allowing users to input their financial information to see predicted scores.

Risks and Mitigations

Risk	Description	Mitigation Strategy
Model Overfitting	Machine learning models, especially complex ones like XGBoost and LightGBM, may learn noise or patterns specific to the training data rather than generalizable trends. This can result in high training accuracy but poor performance on unseen data.	Apply techniques such as cross-validation, early stopping, and regularization (L1/L2). Limit model complexity through hyperparameter tuning and evaluate performance on a separate test dataset.
Feature Bias	Certain features, particularly behavioral indicators, may introduce bias or disproportionately influence the model, leading to unfair or skewed predictions. This is especially relevant when using alternative data sources.	Perform careful feature selection and importance analysis. Use normalization and scaling where appropriate, and assess feature impact using techniques such as SHAP or feature importance scores to ensure balanced representation.
Limited BNPL Research Literature	BNPL is a relatively new financial product, and there is limited academic literature specifically focused on its impact on credit risk. This may restrict theoretical grounding and benchmarking.	Broaden the literature review to include studies on alternative credit scoring, fintech lending, and behavioral finance. Incorporate industry reports and recent working papers to supplement academic sources.

Table 7: Risks and Mitigations

Issues and Resolutions

Issue	Description	Resolution
Lack of Unified Banking Dataset	One of the major challenges was the absence of a comprehensive dataset that captures financial behavior across multiple institutions, including BNPL transactions. Most available datasets are limited to single institutions and do not reflect real-world multi-platform usage.	A multi-institution dataset was developed by integrating real banking transaction data with aggregated BNPL-related data. This approach enabled a more realistic representation of consumer financial behavior across different platforms.
Model Prediction Instability	Initial model outputs showed inconsistency in predicted probabilities, particularly when applied to new or unseen data. This instability can reduce trust in the model and affect decision-making.	Probability calibration techniques such as Platt scaling or isotonic regression were applied to align predicted probabilities with actual outcomes. This improved the reliability and interpretability of model predictions.
Feature Mismatch During Model Deployment	Differences between training data features and deployment input data caused inconsistencies, leading to errors or degraded model performance. This is a common issue when moving from development to real-world application.	A standardized feature schema was implemented to ensure consistency between training and deployment environments. This included defining fixed feature formats, data types, and preprocessing pipelines to maintain uniformity across all stages.

Table 8: Issues and Resolutions

Unified Dataset Challenge

In the real-world financial ecosystem, a consumer will interact with different platforms, including banks, wallets, and BNPL services. The absence of a unified dataset will create challenges in capturing a comprehensive understanding of the financial ecosystem. The construction of a unified dataset will allow the model to tap into a broader set of behavioral patterns, enhancing the predictive capability of the model.

Prediction Stability

In the field of machine learning, there is a risk of a boosting algorithm making unstable predictions. The predictions may be correct in terms of classification, but the probability may not accurately reflect the risk. Calibration is the process of ensuring that, for example, a 70% risk of default is actually close to the actual default rate.

Deployment Consistency

One of the challenges with a typical machine learning project is the transition from development to deployment. Even minor inconsistencies in the feature pipeline will create a significant impact on the overall performance of the deployed model. The standardization of the feature pipeline will ensure the smooth deployment of the model.

Conclusions and Recommendations

6.1 Conclusion

The results obtained from the completed model development and evaluation provide strong empirical evidence supporting the effectiveness of integrating behavioral financial data into credit risk modeling. The study successfully demonstrates that the inclusion of BNPL-related variables and transaction-level behavioral indicators significantly enhances the predictive performance of credit scoring models.

The comparative analysis between baseline models (using only traditional financial variables) and enhanced models (incorporating behavioral features) shows a clear improvement in classification performance. In particular, metrics such as ROC-AUC, recall, and F1-score indicate that models enriched with behavioral variables are more effective in identifying high-risk consumers. This validates the hypothesis that BNPL usage patterns, spending volatility, and liquidity indicators serve as meaningful predictors of credit risk.

The implementation of advanced machine learning algorithm, LightGBM, further reinforces the robustness of the approach. These models effectively captured complex, non-linear relationships within the data, outperforming traditional modeling approaches in distinguishing between defaulters and non-defaulters.

Additionally, the study highlights the importance of alternative data in improving credit access for thin-file and new-to-credit individuals. By leveraging transaction-level data, the proposed framework enables a more inclusive and comprehensive assessment of creditworthiness, addressing a key limitation of conventional credit scoring systems.

Overall, the findings confirm that a BNPL-aware behavioral credit scoring model is not only technically feasible but also practically valuable. The developed FINSCORE framework demonstrates strong potential to enhance credit risk assessment, improve default prediction, and support more informed lending decisions in modern financial ecosystems.

6.2 Recommendations

1. Adoption of Behavioral Credit Scoring in Financial Institutions

Financial institutions should consider integrating behavioral financial indicators, including BNPL usage patterns, spending volatility, and liquidity metrics, into their existing credit risk assessment frameworks. The results of this study demonstrate that such integration significantly improves the detection of high-risk borrowers and enhances overall model accuracy.

2. Implementation of the FINSCORE Framework

The proposed FINSCORE system should be further developed into a deployable scoring framework that combines traditional credit variables with behavioral indicators. This unified scoring approach can provide a more holistic and dynamic view of consumer creditworthiness, particularly in digital-first financial environments.

3. Regulatory Integration and Standardization

Regulators and policymakers should encourage the inclusion of BNPL data within formal credit reporting systems. Standardizing the reporting of BNPL transactions can improve transparency and reduce the risk of hidden debt accumulation, thereby strengthening the overall credit ecosystem.

4. Enhancement of Data Infrastructure

To operationalize such models, financial institutions should invest in data infrastructure capable of aggregating and processing multi-source transaction-level data in real time. This includes integration with open banking systems and Account Aggregator frameworks to enable secure and consent-based data sharing.

5. Focus on Ethical and Explainable AI

Given the use of machine learning models, it is essential to ensure transparency, fairness, and accountability in credit decision-making. Institutions should adopt explainable AI techniques to interpret model outputs and mitigate risks related to bias and lack of transparency.

6. Expansion of Behavioral Feature Space

Future implementations can further enhance model performance by incorporating additional behavioral variables such as:

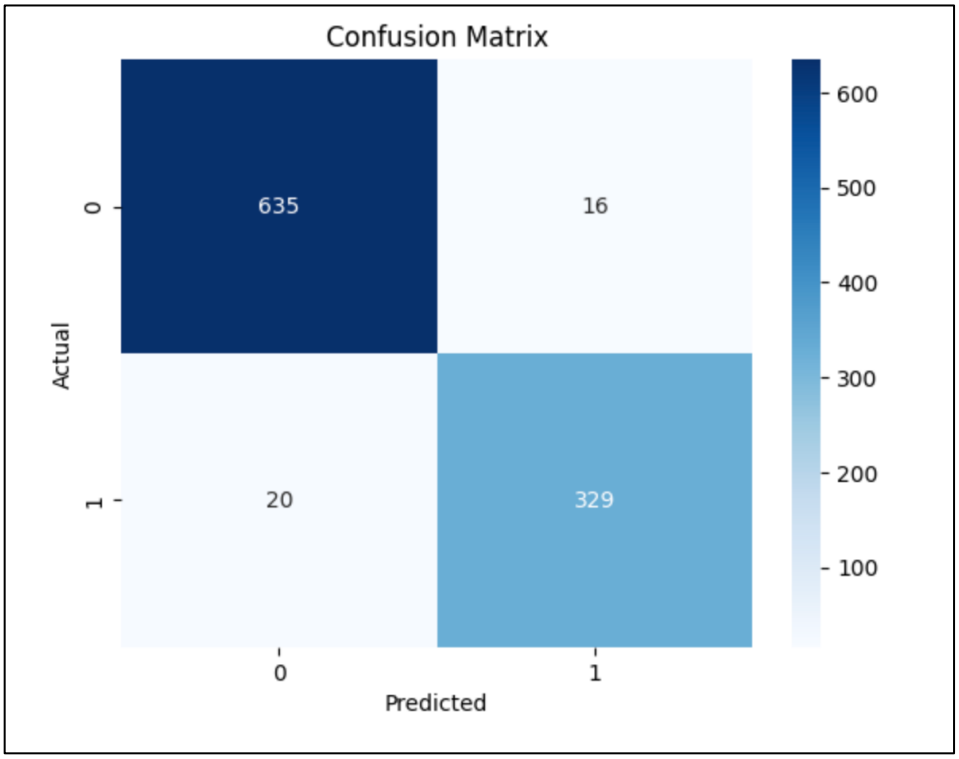
- Temporal spending patterns
- Income consistency indicators
- Payment delay patterns
- Cross-platform credit exposure

This can improve early detection of financial stress and refine risk segmentation.

7. Continuous Model Monitoring and Updating

Since consumer financial behavior is dynamic, models should be periodically retrained and updated using recent data. Continuous monitoring ensures that the model remains accurate and relevant in changing economic conditions.

6.3 Confusion Matrix



ROC AUC: 0.8917033085532947

	Precision	Recall	F1-Score	Support
0	0.97	0.98	0.97	651
1	0.95	0.94	0.95	349
Accuracy			0.96	1000

Table 9: Results

Annexures

Dataset Feature Engineering

bits-project-final-year.ipynb

Users > rajitaroy > projects > finscore > bits-project-final-year.ipynb > # This Python 3 environment comes with many helpful analytics libraries installed

Generate | Code | Markdown | Run All | Clear All Outputs | Outline | ... | Select Kernel

```python
# ----------------------------
# CASH ADVANCE FEATURES
# ----------------------------

cash_txn = transactions[transactions["channel"] == "credit_cash"]

cash_features = cash_txn.groupby("customer_id").agg(
    credit_cash_transactions=("amount","count"),
    credit_cash_total=("amount","sum")
).reset_index()

dataset = dataset.merge(cash_features, on="customer_id", how="left")

dataset[["credit_cash_transactions","credit_cash_total"]] = \
dataset[["credit_cash_transactions","credit_cash_total"]].fillna(0)
```

[181] Python

```python
# ----------------------------
# EXTERNAL SUPPORT FEATURES
# ----------------------------

transfer_txn = transactions[transactions["channel"] == "transfer_in"]

transfer_features = transfer_txn.groupby("customer_id").agg(
    external_support_transactions=("amount","count"),
    external_support_total=("amount","sum")
).reset_index()

dataset = dataset.merge(transfer_features, on="customer_id", how="left")

dataset[["external_support_transactions","external_support_total"]] = \
dataset[["external_support_transactions","external_support_total"]].fillna(0)
```

[182] Python

```python
dataset["spend_income_ratio"] = dataset["total_spend"]/dataset["monthly_income"]

dataset["bnpl_usage_ratio"] = dataset["bnpl_transactions"]/dataset["total_spend"]

dataset["bnpl_income_ratio"] = dataset["bnpl_total"]/dataset["monthly_income"]
```

```python
def classify_customer(row):

    if row["bnpl_income_ratio"] > 0.35:
        return "bnpl_dependent"

    elif row["credit_utilisation"] > 0.75:
        return "cc_revolver"

    elif row["spend_income_ratio"] > 1.2:
        return "liquidity_stressed"

    elif row["spend_volatility"] > row["avg_spend"]:
        return "volatile_spender"

    elif row["credit_utilisation"] < 0.4 and row["bnpl_income_ratio"] < 0.1:
        return "financially_stable"

    else:
        return "mixed_behavior"


dataset["behavior_segment"] = dataset.apply(classify_customer,axis=1)
```

[193] Python

```python
dataset = pd.get_dummies(
dataset,
columns=["employment_type","city_tier","behavior_segment"],
# columns = ["behavior_segment"],
drop_first=True
)
```

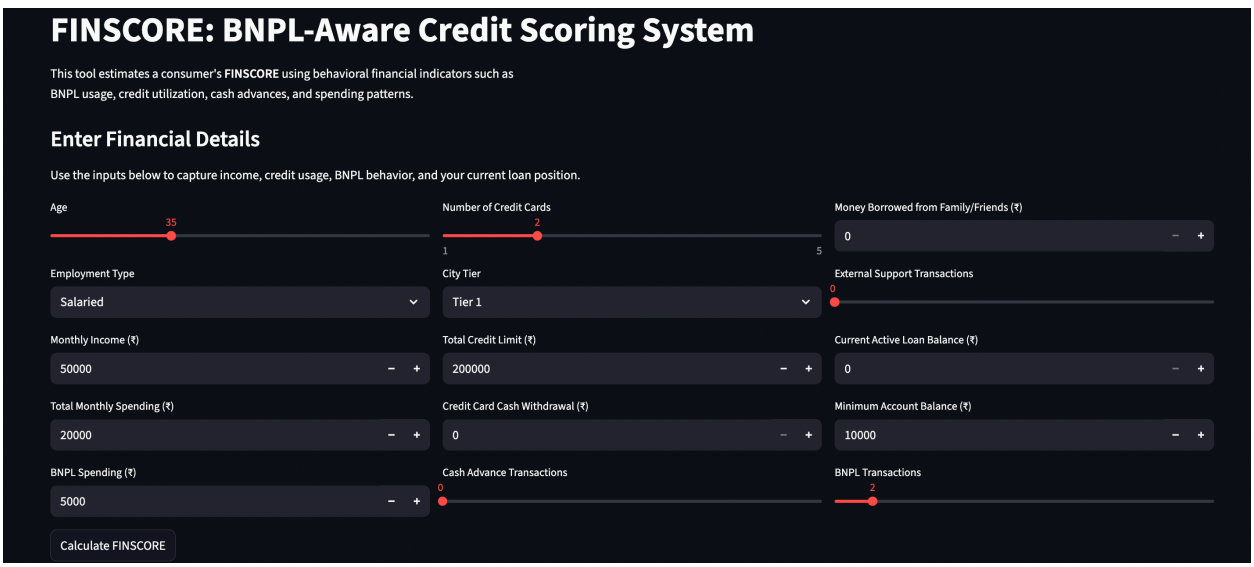
[194] Python

Generate | Code | Markdown

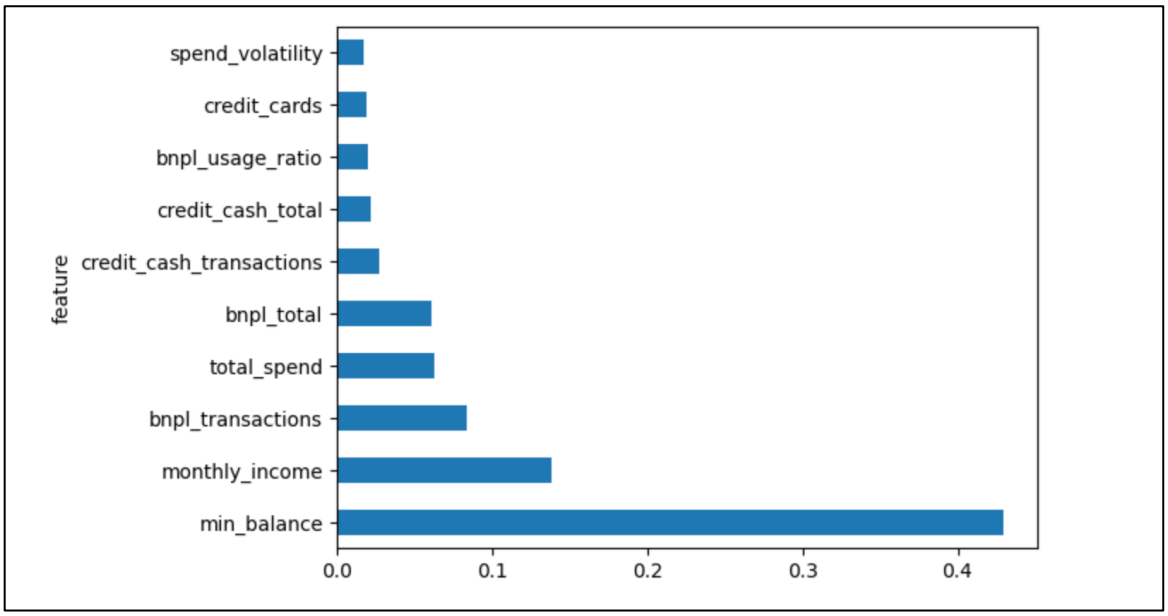
```python
from sklearn.model_selection import train_test_split

features = [
"age",
"monthly_income",
"credit_cards",
"credit_limit_total",
"total_spend",
"avg_spend",
"spend_volatility",
"max_spend",
"bnpl_total",
"bnpl_transactions",
"min_balance",
"bnpl_usage_ratio",
"employment_type_self_employed",
"city_tier_tier2",
"city_tier_tier3",
"credit_cash_total",
"credit_cash_transactions",
"external_support_total",
"external_support_transactions",
"cash_advance_ratio",
"external_support_ratio",
"segment_cash_dependent",
"segment_external_support"
]

X = dataset[features]
y = dataset["default"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    stratify=y,
    random_state=42
)
```

[195] Python

Streamlit Dashboard



Model Feature Importance



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2. PwC India (2022). *The Growth of BNPL in India: Opportunities and Risks for Financial Institutions*. PricewaterhouseCoopers India.

Glossary

Term	Definition
BNPL (Buy Now Pay Later)	A short-term financing option that allows consumers to make purchases and pay in installments over time, often without interest.
Credit Scoring	A statistical method used to evaluate the creditworthiness of a borrower based on financial data.
FINSCORE	A proposed behavioral credit scoring framework developed in this study using transaction-level and BNPL data.
Credit Utilization	The ratio of total credit used to total available credit limit.
Spend-to-Income Ratio	A measure indicating the proportion of income spent by a consumer.
BNPL Usage Ratio	The proportion of total spending made through BNPL transactions.
Liquidity	The ability of a consumer to meet short-term financial obligations.
Liquidity Stress	A condition where a consumer frequently experiences low or negative account balances.
Transaction-Level Data	Detailed records of individual financial transactions including date, amount, and category.
Alternative Data	Non-traditional financial data used for credit assessment, such as transaction history and digital payments.
Behavioral Credit Scoring	A credit evaluation approach based on financial behavior patterns rather than only historical credit records.

Machine Learning	A set of algorithms that enable systems to learn patterns from data and make predictions.
Logistic Regression	A statistical model used for binary classification problems such as default prediction.
Random Forest	An ensemble machine learning model that uses multiple decision trees for prediction.
XGBoost	An advanced gradient boosting algorithm widely used for predictive modeling.
LightGBM	A gradient boosting framework optimized for speed and efficiency in large datasets.
ROC-AUC	A performance metric that measures a model's ability to distinguish between classes.
Precision	The proportion of correctly predicted positive observations out of total predicted positives.
Recall	The proportion of correctly predicted positives out of actual positives.
F1-Score	A harmonic mean of precision and recall used to evaluate model performance.
Open Banking	A system that allows financial institutions to share customer data securely across platforms with consent.
Account Aggregator	A framework that enables consolidation of financial data from multiple institutions into a unified view.
Feature Engineering	The process of creating new variables from raw data to improve model performance.

Summary of how the feedback for Project Outline and Mid-Semester Project Report have been addressed

Mid-semester:

The feedback provided by the faculty examiner in respect to the initial project outline has been reviewed and incorporated into a revised project framework.

Firstly, a change in the project framework from a conceptual demonstration to an empirical research study has been made and reflected in a revised title for the project: "An Empirical Evaluation of Buy Now Pay Later (BNPL) – Integrated Behavioral Credit Scoring Using Unified Multi-Institution Transaction Data." Research questions have been incorporated into the revised project framework to provide a basis for conducting the study and have been designed to evaluate the impact of BNPL and transactional data in respect to credit risk prediction.

Secondly, a revision in respect to the project objectives has been made by consolidating them for greater clarity and understanding. A significant revision in respect to methodology has been made by incorporating a description of the research design, sample framework, and data generation methodology, and a description of behavioral feature engineering and model development.

Thirdly, the study now includes particular machine learning models such as, XGBoost, LightGBM, etc., together with a framework of evaluation using performance metrics such as Accuracy,

Precision, Recall, F1 Score, ROC AUC, etc. This satisfies the need for empirical testing.

Finally, the literature review section of the project now includes BNPL, alternative data credit scoring, behavioral finance, machine learning credit scoring, etc., with appropriate academic literature references using APA Style.

Thus, the revised project shows improvements in academic rigor, clarity, and conformity with current trends in FinTech research.

Final report:

The feedback provided after the mid-semester evaluation has been carefully reviewed and incorporated into the final dissertation, resulting in significant improvements in analytical depth, practical relevance, and academic rigor.

Firstly, the literature review has been strengthened by incorporating industry statistics and real-world case studies from global BNPL markets such as the United States and Australia. This enhances the contextual understanding of BNPL growth and highlights the urgency of integrating such data into credit risk assessment frameworks.

Secondly, a detailed cost-benefit analysis has been introduced to evaluate the practical feasibility of implementing the proposed behavioral credit scoring model. This includes estimation of development costs, infrastructure requirements, and potential financial benefits such as reduction in default rates and improved risk assessment efficiency. Additionally, relevant banking regulatory considerations have been examined to assess compliance requirements and integration challenges.

Thirdly, the study now includes an evaluation of long-term economic viability, focusing on scalability, sustainability, and the potential impact of adopting behavioral credit scoring systems within financial institutions.

Further, operational and regulatory challenges associated with real-world implementation have been explicitly identified. These include issues such as data privacy concerns, integration with existing banking systems, lack of standardized BNPL reporting, and model interpretability requirements.

Moreover, the dissertation has incorporated interdisciplinary perspectives by linking behavioral finance concepts with data-driven credit modeling, thereby providing a more comprehensive understanding of consumer financial behavior.

Finally, the study has expanded its academic depth by including a more detailed discussion on ethical considerations, including bias in machine learning models, transparency, and responsible use of alternative data.

Overall, the final report reflects a transition from a technically sound model to a well-rounded, industry-relevant research study that addresses practical, regulatory, and ethical dimensions of BNPL-integrated credit scoring.

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Semester	Final semester
Project Title	Design and Evaluation of an Al-Based Behavioral and BNPL-Aware Credit Risk Prediction Model Using Transaction Data

Particulars of the Supervisor and Additional Examiner

	Supervisor	**Additional Examiner**
Name	Raunak Rhishabh	Rameshwar Shelge
Qualification	BTech in Electronics and Communications	BE, Computer Science & Engineering
Designation	Lead Engineer	Founder and CEO
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Remarks of the Supervisor on Final Project Report

The project outline is comprehensive and covers the necessary elements for the upcoming phases. The chosen topic is contemporary and aligns well with current industry trends. I look forward to seeing the progress in subsequent stages of the project.

	Raunak	Ram Shelge
Signature of Student	**Signature of Supervisor**	**Signature of Additional Examiner**
Name: Harshit Derashree	**Name: Raunak Rhishabh**	**Name: Rameshwar Shelge**